

Michael Ma

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Education

The University of Illinois at Urbana-Champaign

December 2023

B.S in Computer Science

GPA: 3.7/4.0

Experience

Basis Technologies, Chicago IL — Software Engineer (January 2024 - Present)

- Working on creating automated regression tests for the DSP backend in Java
- Created internal tools to make daily work more efficient
- Knowledgeable in ad-tech and inner workings of DSP
- Worked with SQL, Kubernetes, MongoDB, Clickhouse, Redis to manipulate and store data

Nokia, Naperville IL — Software Engineering Co-op (September 2023 - Dec 2023)

- Worked on internal search engine using AI, machine learning and NLP
- Trained models using numerous datasets to optimize search engine results in Python
- Worked on the front end of the search engine using Typescript

Basis Technologies, Chicago IL — Software Engineering Intern (June 2023 - August 2023)

- Improved internal testing tool to automate log checking for errors/warnings for components
- Created automated regression tests for multiple DSP components

Lockstep, Seattle WA — Software Engineering Intern (January 2022 - August 2022)

- Worked on implementing and optimizing inputting new API calls into Azure Table Storage
- Improved the Xero and QBO connectors by adding sync functionality.
- Worked on a team of two to build a MVP of a connector to Netsuite to sync data from Netsuite and insert it into Lockstep SQL tables.

University of Illinois at Urbana Champaign, Urbana IL – Course Assistant

- CS222: Software Design Lab (January 2022 - May 2022) - Mentored a group of 5 students in creating a semester-long project in Java, C++. Attended weekly meetings and graded progress.
- CS225: Data Structures (August 2021 - December 2021) - Held office hours and exam review to help students debug and understand how to resolve issues.
- CS126: Software Design Studio (January 2021 - May 2021) - Held office hours and moderated code review, emphasized writing readable and extendable code. Graded weekly projects.

Nobel Coaching and Tutoring, North Potomac MD – Software Engineering Intern

(May 2019 - August 2019)

- Designed and created an online web game for elementary schoolers to teach soft skills such as teamwork and communication. Used Node.JS, JavaScript, socket.IO to build backend.

Projects

Personal Website (April 2024)

- Created a personal website of pure awesomeness
- Used modern day web design principles to make it as beautiful as possible

Delaunay Triangulations and Voronoi diagrams (Spring 2022)

- Created a program to efficiently triangulate a set of random points in loglinear time using the Bowyer-Watson algorithm.
- Implemented turning the triangulation into a Voronoi diagram in linear time

N-Body Simulation (Winter 2021)

- Created a simulation of a galaxy with ~40k stars using an efficient algorithm in loglinear time called Barnes-Hut that uses quadtrees (octree for three dimensional simulations)
- Simulated a black hole and the movements of stars around black hole

Skills

Languages: Python, C++, C, Java, TypeScript, HTML, CSS

Machine Learning: PyTorch, TensorFlow, OpenCV, Keras

Databases: SQL, Clickhouse, Redis, MongoDB, Azure Table Storage

Others: Jenkins, Kubernetes, GCP, Git, Docker, Shell